

Python Programming – Study Notes

1. What is Python?

Python is a high-level, interpreted programming language known for its simple syntax and readability. It is widely used in web development, data science, artificial intelligence, automation, and education.

2. Variables

A variable is a name used to store a value. Examples include `name = 'Mahesh'` and `age = 20`. Python determines the data type automatically.

3. Data Types

Common Python data types include `int` for integers, `float` for decimal numbers, `str` for text, `bool` for True or False values, `list` for ordered collections, and `dict` for key-value data.

4. Conditional Statements

The `if`, `elif`, and `else` statements allow a program to make decisions based on conditions.

5. Loops

The `for` loop is commonly used to repeat an operation over a sequence. The `while` loop repeats while a condition remains true.

6. Functions

A function is a reusable block of code. Functions are defined using the `def` keyword. They can accept parameters and return values.

7. Lists

A list stores multiple values in a single variable. Example: `fruits = ['apple', 'banana', 'mango']`. Lists are mutable, so their contents can be changed.

8. Object-Oriented Programming

Object-oriented programming organizes software using classes and objects. Important concepts include encapsulation, inheritance, polymorphism, and abstraction.

9. Python and Artificial Intelligence

Python is popular in AI because it has libraries and frameworks for machine learning, data processing, natural language processing, and deep learning.